

AgroStruxure™

Solar-synchronised agri-energy microgrid and precision irrigation for PM-KUSUM pumps

“One solar pump can become a rural energy node.”

A retrofit kit that caps irrigation at what the crop needs, so free daytime solar cannot over-pump the aquifer, then runs a shared farm-gate pre-cooler from the idle surplus.

41.2%

less groundwater

7,000 m³/ha/yr

63%

of PV output idle

after right-sizing water

1.31 t

tomatoes saved

per ha per year

2.0 yr

capital payback

₹50,212 net per farm

Sanskar Tiwari (Firmware, lead) · Shambhavi Patil (Geospatial) · Kanishka Salgude (Power / VFD) · Chaitanya Ranade (Agronomy / Product)

Model: research/08_IMPACT/impact_model.py (1 ha tomato, Nashik, 4.8 kWp PM-KUSUM Component B pump). 63% idle PV = after irrigation is right-sized (37% idle today).

ONE PUMP · TWO LOADS · ONE CONTROLLER



4.8 kWp PV array

6,559 kWh per year



AgroStruxure edge controller

FAO-56 budget · field-capacity cutoff · TeSys changeover



Right-sized drip

Altivar ATV320 pump
2,422 kWh/yr



Shared pre-cooler

4-farm cluster
1,062 kWh/yr

–41.2% groundwater

–52% farm-gate loss

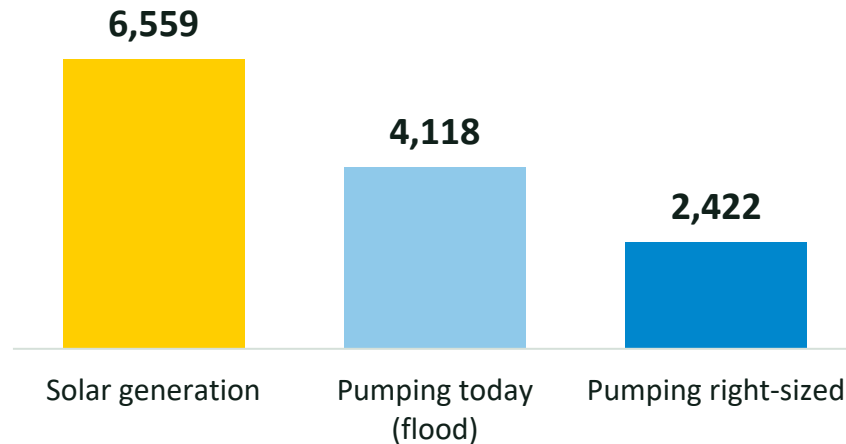
Altivar Solar ATV320 · TeSys D · Modbus RTU · EcoStruxure-aligned

Free Solar Power Accelerates Groundwater Collapse

Under PM-KUSUM Component B, daytime pumping costs nothing, so nothing limits the water drawn, and the sun goes unused.

Annual energy, one 4.8 kWp pump

1 ha tomato, kWh per year



Today: 2,442 kWh idle (37%).

Right-sized: 4,137 kWh idle (63%), because the pump now needs 1,696 kWh less.

01

Free power, unlimited pumping

Flood irrigation applies 17,000 m³/ha/yr against a 10,000 m³ crop entitlement. Zero marginal cost removes any reason to stop.

16–39%

more water pumped after solarisation (Rajasthan study, IWMI)

02

Solar capacity stranded

Once the water is right-sized, most of the array's output has no pumping use: a stranded asset at midday.

63%

of PV output idle after right-sizing water

03

Harvest rots where the power is

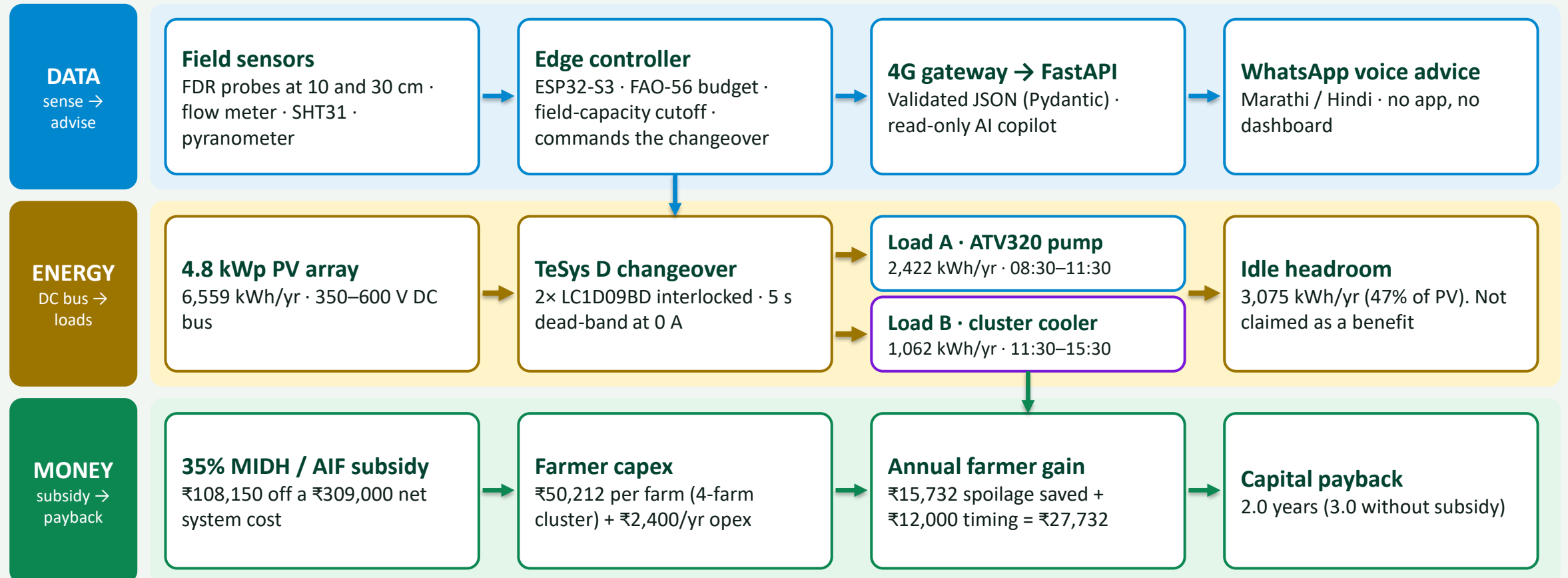
No cold chain at the farm gate: tomatoes are lost at farm stage (11.62% including market handling).

8.37%

farm-stage tomato loss (NABCONS 2022, MoFPI)

One Retrofit Kit, Three Flows: Data, Energy, Money

Sensing, switching and settlement in one design. The controller works offline; the cloud only advises.



41.2% Less Groundwater, Full Crop Requirement Met

A dual-Kc FAO-56 budget sets the entitlement; dual-depth FDR probes stop the pump at field capacity.

Dual-depth soil moisture profile

0–10 cm · surface probe

Evaporative zone, typically 18–32% VWC. Tracks diurnal loss.

10–40 cm · root-zone probe

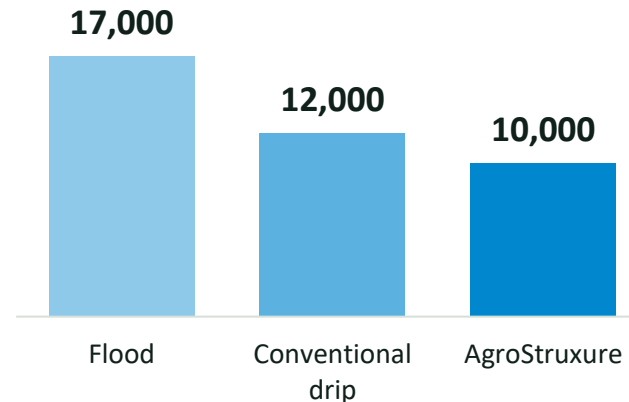
Irrigation stops at field capacity (45% VWC), so there is no root-zone hypoxia.

40 cm + · deep percolation

Pulsed drip leaves nothing to leach below the root zone.

Cutoffs are calibrated per soil type.

Water applied, m³ per hectare per year



7,000 m³

conserved per ha per year

5,000 m³ from drip hardware

2,000 m³ from the entitlement cap,
locked against rebound

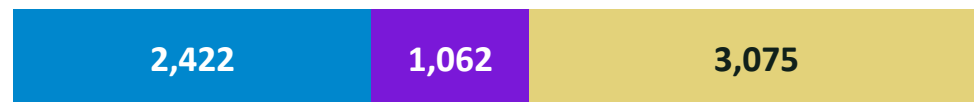
Per hectare per year	Flood	AgroStruxure	Change
Water applied	17,000 m ³	10,000 m ³	–7,000 m ³
Pumping electricity	4,118 kWh	2,422 kWh	–1,696 kWh

Assumes drip is already installed (existing or subsidised). Drip hardware is not in the capex.

Cooling a 4-Farm Cluster Uses 26% of One Farm's Idle Solar

Power is switched upstream on the DC bus at zero current, so no inductive spike reaches the drive.

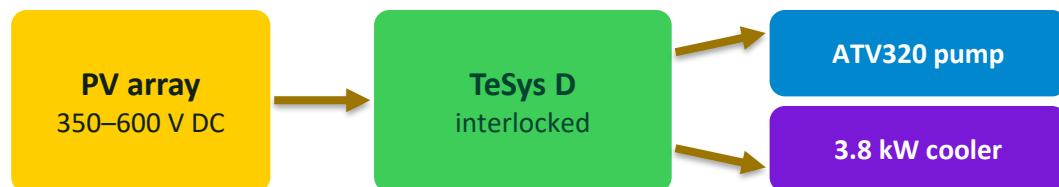
Where the host farm's 6,559 kWh/yr goes (kWh)



■ Pumping 37% ■ Cluster cold chain 16% ■ Headroom 47%

1,062 kWh = 26% of the 4,137 kWh surplus. 60 batches a year (4 farms × 15) at 17.7 kWh each. Headroom is reported, not claimed.

Single-line diagram



Five-step zero-current transfer

- 1 Saturation or quota reached**
30 cm probe hits 45% field capacity, or the daily entitlement is used. **T = 0 s**
- 2 Controlled VFD ramp-down**
Modbus Reg 8502 takes the ATV320 to 0 Hz over 8 s. **0 → 8 s**
- 3 Zero-flow valve closure**
Latching solenoid shuts at zero velocity: no water hammer. **T = 8 s**
- 4 5 s DC dead-band**
Bus capacitors discharge and DC current reaches 0 A. **8 → 13 s**
- 5 TeSys D transfer**
Interlocked contactor energises the 3.8 kW compressor. **T = 13.2 s**

Halving Farm-Gate Tomato Spoilage: 8.37% to 4.00%

A shared 2 MT pre-cooler pulls harvest-hot tomatoes from 32 °C to a safe 12 °C within four hours.



Illustrative image

1.31 t

tomatoes saved per ha per year (2.51 t
→ 1.20 t)

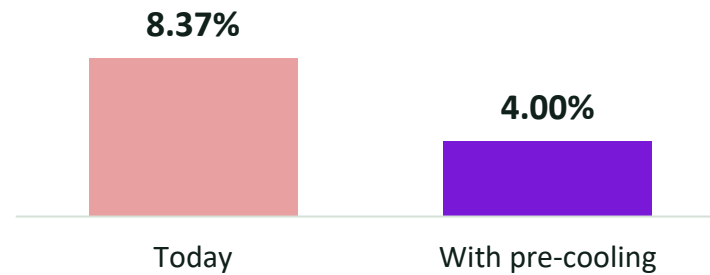
₹15,732

spoilage saved per year at ₹12/kg

₹12,000

price timing: hold crates for the evening
mandi (+₹1–2/kg on 8 t)

Farm-stage loss, % of harvest



Thermal physics

41 kWh of heat removed per 2 MT batch (32 → 12 °C).

13.7 kWh of electricity at COP 3.0: 3.43 kW for 4 h, within the 4.8 kWp array.

12 °C setpoint avoids chilling injury below 10 °C. Phase-change plates hold overnight.

Net farmer gain ₹25,332 per year = ₹15,732 + ₹12,000 – ₹2,400 opex

Embodied emissions avoided: 0.39 t CO₂e per ha per year

Vernacular Advice Grounded in Hardware Truth

Spoken Marathi and Hindi advisories on WhatsApp. The AI can read telemetry; it can never switch a contactor.

Four-stage trust pipeline

- 1 Edge telemetry**
MCU logs FDR moisture, flow and VFD status over Modbus RTU.
- 2 Schema validation**
Pydantic checks every JSON record; bad data blocks the model.
- 3 Read-only RAG prompt**
Copilot sees telemetry and agronomy templates, with no tool access.
- 4 Voice synthesis**
Marathi / Hindi audio delivered as a WhatsApp voice note.

```
{"vwc_30cm": 0.448, "status": "FC_CUTOFF", "pv_kw": 4.21, "load": "PRE_COOLER"}
```

Hard invariant: only the FreeRTOS state machine switches power; a manual IP67 switch overrides.

K

Krishi Mitra

WhatsApp voice advisory · Nashik cluster

रामभाऊ, तुमच्या टोमॅटोच्या मुळांना पुरेसे पाणी मिळाले आहे (४४.८%). पंप सुरक्षितपणे बंद झाला असून सौर वीज शीतगृहाकडे वळवली आहे.



0:14 / 0:26

Translation: "Rambhau, your tomato roots have reached optimal moisture (44.8%). Pumping stopped safely; solar power now runs the cold room."

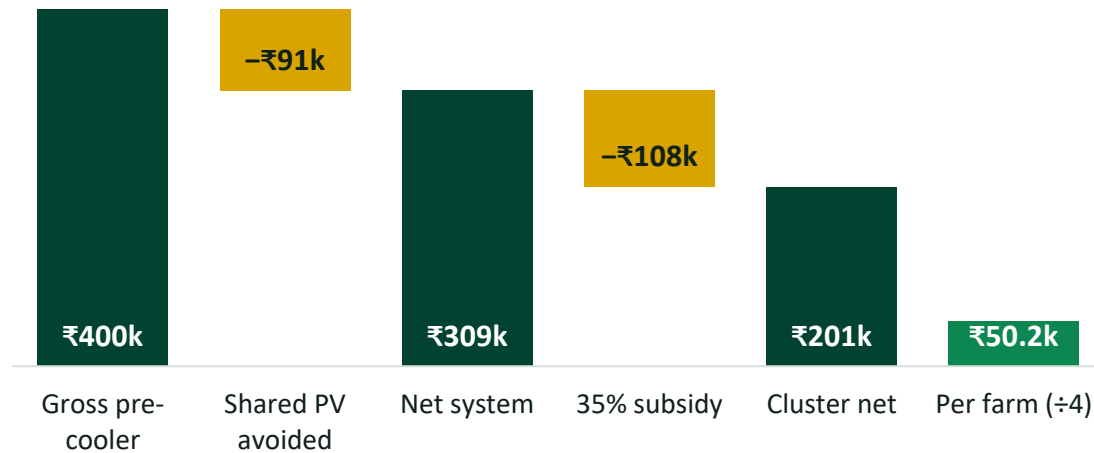
धन्यवाद! उद्या पाणी कधी सुरू होईल?

UX artifact: 20-second voice notes at key state changes (pumping done, cold room online); built for zero-literacy use.

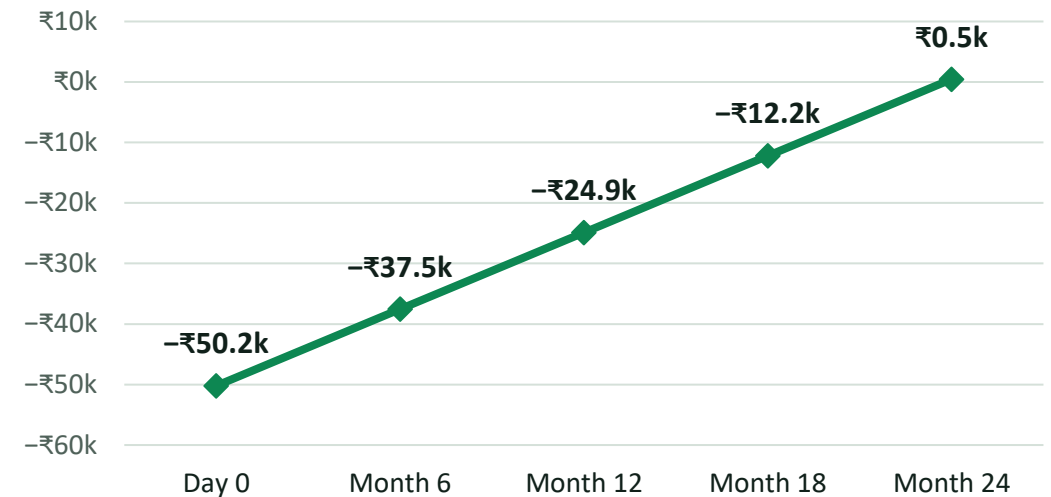
₹50,212 Net Capex per Farm, Payback in 2.0 Years

A farm's share of a 4-farm pre-cooler after the 35% subsidy. The ₹7,540 edge controller is separate and pays back in 1.5 years.

Capex waterfall, ₹ thousand per 4-farm cluster



Cumulative cash flow per farm, ₹ thousand



Annual farmer gain

₹25,332

₹15,732 spoilage + ₹12,000 timing – ₹2,400 opex

Edge controller BoM

₹7,540

Pays back in 1.5 yr; ≈1.9 yr combined with the cooler share

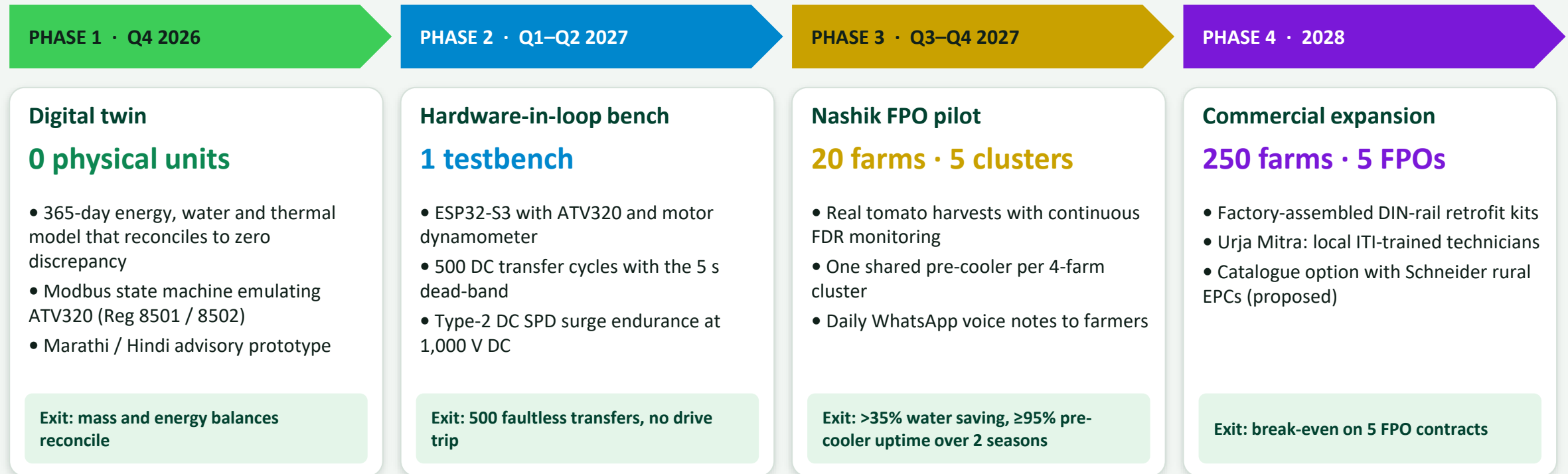
Tier B FPO hub (5 MT)

4.2 yr

Own array, leasing: ₹780k net capex, ₹184k/yr net revenue

From Digital Twin to a 250-Farm Pilot

Each phase has an exit gate. Targets below are plans, not results.



Enablers PM-KUSUM Component B (~10.06 lakh pumps installed, 4.67 lakh in Maharashtra) · Atal Bhujal Yojana · MIDH / AIF 35%. **Proposed partners (not yet confirmed):** Sahyadri Farmers Producer Co., Nashik; Schneider Electric rural EPC channel.

One Solar Pump. Four Measured Outcomes.

A connected retrofit layer for Altivar Solar drives and TeSys switchgear that turns standalone pumps into community microgrids.

41.2%

LESS GROUNDWATER

7,000 m³ per ha per year; full crop requirement still met

4,137 kWh

IDLE SOLAR UNLOCKED

per farm per year once water is right-sized; cooling a 4-farm cluster uses 1,062 kWh

1.31 t

TOMATOES SAVED

per ha per year; farm-stage loss 8.37% → 4.00%

2.0 years

CAPITAL PAYBACK

₹50,212 net capex per farm;
₹25,332 net gain per year

Sustainability 0.39 t CO₂e per ha per year (embodied emissions of avoided spoilage) · 7,000 m³ groundwater conserved · no diesel, no battery

Sanskar Tiwari

Embedded firmware and lead

FreeRTOS state machine, Modbus RTU, CiA402 drive control

Shambhavi Patil

Geospatial intelligence

Sentinel-2 NDVI/NDWI, dynamic FAO-56 Kc curves, aquifer balance

Kanishka Salgude

Electrical systems and VFD

Altivar ATV320 configuration, TeSys D interlocking, DC routing

Chaitanya Ranade

Product and agronomy

FPO cluster economics, MIDH/AIF subsidies, vernacular WhatsApp UX